



Review

Energy poverty and beyond: The state, contexts, and trajectories of energy poverty studies in Asia

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ABSTRACT

There is a dichotomy in the understanding of energy poverty between Asia and the rest of the world, especially in the Global North. While there has been considerable discourse regarding energy poverty in the Global North, particularly the emergence of affordable cooling, the development of energy poverty studies in the Global South has received less attention. This review article explores energy poverty within Asia reviewing energy poverty studies in 23 countries and territories in South, East, and Southeast Asia, using bibliometric analysis to identify emerging discussions and researchers. We highlight the concepts of 'hidden energy poverty', 'urban energy poverty', and 'cooling poverty' to outline potential areas for future research that contextualises energy poverty into using a more encompassing term, 'energy insecurity,' which consolidates the wide gamut of energy-related sufferings. To investigate other drivers and impacts of energy insecurity, qualitative research can extend previously identified elements of accessibility and affordability. New research directions may also include contextualising energy insecurity in the context of a just energy transition.

1. Introduction

There is a stark contrast in the perception of energy poverty between Asia and other world regions, particularly in the Global North. For instance, energy poverty in the Global South is often, and mistakenly, perceived solely as a problem of limited access to clean fuels. In contrast, in the Global North, it primarily arises from the unaffordability of modern fuels. Given that Asia is a continent with countries that belong to both the Global North and the Global South, research on energy poverty in the continent may offer potential ways to reconcile this accessibility-affordability divide. Furthermore, as a continent undergoing momentous change and growth, understanding and measuring energy poverty in Asia is crucial for future discussions on energy poverty worldwide. Thus, this paper presents a review of the extant literature on energy poverty in South, East, and Southeast Asia. In so doing, we identify the different perceptions, measurements, determinants, impacts, and emerging discussions of energy poverty in these regions.

1.1. Energy poverty in the Global North

Energy poverty, defined as 'the inability of a household to access socially and materially necessitated levels of energy services in the

home' [1], is conventionally traced back to Boardman's [2] seminal work. In this book, she termed the issue 'fuel poverty,' following Lewis's earlier definition of 'the inability to afford adequate warmth in the home' [3]. Boardman methodologically presented the issue and defined a quantitative indicator for this specific form of deprivation. She found that energy-poor households in the United Kingdom (UK) spent around 5% of their income on energy, twice the median household expenditure on energy at that time. She succinctly formulated that households were energy-poor if they spent more than 10% of their income on energy expenditures, leading to the commonly known "10% indicator". However, the simplicity of this indicator caused 'energy poverty' to be conflated with households spending more than 10% of their income on energy expenditures rather than as the challenge of the inaccessibility of energy services.

While Boardman used the term 'fuel poverty,' other researchers studying similarly developed countries have adopted 'energy burden' and 'energy poverty' to describe the same phenomenon [4–6]. However, in developing countries, the term 'energy poverty' primarily revolves around the accessibility of clean fuels rather than the affordability of energy. To ensure clarity, we use the term 'energy poverty' in this paper to mean a household's lack of access to energy access.

Energy poverty in the Global North has been the subject of much

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research, particularly in regions such as the EU [1,7–9], and countries like France [10], Greece [11,12], Portugal [13], Spain [14,15], Australia [16], the US [17,18], and the UK [19,20]. Most of this research has focused on the households' inability to afford heat in cold weather and has aimed to identify the drivers of energy poverty, the affected groups, the consequences of this deprivation on affected households, and its extent within a region or a country. The elderly and pensioners [8,10,21,22], renters [8,10,15,21], those living in rural areas [10,21,22], those living alone [10,21], ethnic minorities [18,23], and those living on low incomes [11,13] are among the groups identified as more susceptible to energy poverty. Martiskainen et al. [24] have also identified those living with health and mobility difficulties, and those living in an older or uninsulated property, as more vulnerable to energy poverty. Health effects associated with energy poverty include negatively impacted subjective well-being [16], excess winter morbidity among the elderly [25], decreased mental health due to stress associated with affording energy [26], and respiratory ill-health due to the presence of damp and mold, resulting from the household's inability to afford adequate energy services [19].

Most studies on energy poverty in the Global North have utilised Boardman's 10% indicator in national household surveys. However, while this measurement has its advantages, it also has several issues. Firstly, the 10% indicator is an absolute threshold and often lacks relevance as it does not consider the median household energy expenditure of the adopting nation [20,27]. Nathan [28] argues that the economic parameters within the indicator, such as income and energy price, are continuous variables, while the 10% indicator makes a discrete distinction between the energy-poor and non-poor households. In practice, there is not much difference between a household spending 9.9% of its income on energy expenditure and another spending 10.1%; yet only the former household will be eligible for assistance from a policy standpoint. The second issue with the 10% indicator is that its absolute value was derived from data taken in 1988, making it highly outdated [20]. Lastly, analysing macro-level data without reaching out to individual households reduces the complex issue of energy poverty into a single dimension of economics. This approach disregards factors such as dwelling conditions and household behaviours that reduce energy expenditure to below 10% of the household income.

In an effort to develop more comprehensive indicators of energy poverty, some researchers have incorporated households' self-assessments. For example, Healy and Clinch [8] used both qualitative subjective and objective indicators, such as the inability to pay utility bills and the lack of central heating, respectively, to supplement the 10% quantitative indicator. Hills [29] proposed the low-income high costs (LIHC) indicator, which defines households as fuel-poor if their income is low (i.e., they fall below the poverty line after accounting for housing costs) and their required energy spending to achieve an adequate standard of warmth is above a certain threshold. Others have proposed multidimensional energy poverty (MEP) indices. For instance, Bouzarovski and Herrero [7] established a cross-country comparative index that accounts for a household's inability to keep their home adequately warm (inability), their arrears in utility bills (arrears), and housing faults such as a leaking roof (housing faults), in the ratio of 2:1:1. Similarly, Sokołowski et al.'s multidimensional index [22] combined two objective indicators, i.e., LIHC and high actual costs, and three subjective indicators, i.e., not warm enough, housing faults, and bill difficulties, classifying a household as energy-poor if they ranked positively for at least two of these indicators. The inclusion of housing faults in Bouzarovski and Herrero's and Sokołowski et al.'s indicators recognises energy poverty as a challenge that extends beyond economic affordability, making it a matter of environmental factors, such as dwelling characteristics. However, these indicators still treat energy poverty as a state that households are in, and do not fully account for household vulnerabilities that increase their risk of experiencing energy poverty.

1.2. Energy poverty in the Global South

As demonstrated, discussion on energy poverty in the Global North have rapidly proliferated in a synchronised manner, primarily focusing on the affordability of energy, with emphasis on providing warmth. This perspective of energy poverty in the Global North diverges from the understanding of energy poverty in the Global South, which places greater emphasis on accessibility, particularly in rural or developing regions where households lack access to modern energy services. Recently, a noteworthy development in the discussion of energy poverty in the Global North has emerged, with a focus on affordable cooling [13,30]. However, studies investigating affordable cooling in the region have only recently begun to emerge and we believe this is significant in light of the expected increase in frequency and intensity of hotter days and nights due to accelerating climate change. Nonetheless, researchers need to consider the Global South, as many countries and regions with warmer, tropical climates are located in this area.

Within the context of energy poverty in the Global South, households are characterised by a lack of access to clean and modern fuels. To measure energy poverty in this context, Nussbaumer et al.'s [31] multidimensional energy poverty index (MEPI) has gained popularity. The MEPI considers access to modern energy services for various purposes, including cooking, lighting, entertainment, education, and communication. This expansion of energy poverty indicators to encompass a broader discussion aligns with the development of indicators in the Global North to increasingly encompass factors outside of income. However, it remains unclear whether discussions of energy poverty in the Global South also incorporate factors such as extreme weather, rapid urbanisation, and geographical and climatic conditions. Additionally, it is uncertain whether energy poverty is being addressed with energy transitions and the pursuit of justice.

Thus, we turn our attention to Asia, a continent with diverse countries and territories to investigate the state of energy poverty discussions. Our review of energy poverty research in Asia reveals that studies are increasingly being contextualised within their respective study areas. However, developments are not uniform across the continent. While new understandings and indicators are being proposed to address the many factors contributing to energy poverty within the region, we find that there is insufficient consideration of energy poverty in the context of extreme weather brought on by climate change, rapid urbanisation, and economic growth. Research is also scarce in assessing energy poverty in urban areas, particularly in relation to the need for cooling.

We begin by describing how we collected the relevant literature (Section 2). We then present a bibliometric study that provides an overview of the current perceptions of energy-related deprivations in various Asian regions and countries (Section 3) and present our in-depth analysis of the definition and measurements/indicators of energy poverty, factors that affect and the impacts of energy poverty, and proposed solutions to address this challenge (Section 4). In this article, we use the term 'energy poverty' to refer to the deprivation of energy, such as the absence of access to modern energy services or situations in which households have higher-than-median energy expenditures. We conclude by discussing the limitations of our study and proposing directions for future research (Section 5).

2. Methods

We conducted a literature review by searching the following databases: Science Direct, JSTOR, Project Muse, and Google Scholar. Science Direct is an Elsevier database that hosts over nineteen million articles and chapters related to physical, social, and health sciences [32]. JSTOR provides access to over twelve million journal articles related to social sciences, science and technology, and arts and humanities [33]. Project Muse is a database that contains humanities and social science books and journals [34]. We also used Google Scholar to search for any

relevant articles that were not indexed in the other databases.

We used the following search terms: ‘energy poverty’, ‘fuel poverty’, ‘energy insecurity’, and ‘energy burden’, combined with 23 countries and territories in South, East, and Southeast Asia, including China, India, Indonesia, Pakistan, Bangladesh, Japan, Philippines, Vietnam, Thailand, Myanmar (Burma), South Korea, Malaysia, Nepal, Sri Lanka, Cambodia, Laos, Singapore, Mongolia, Bhutan, Maldives, Taiwan, Macau (Macao), and Hong Kong. We also included the keyword ‘Asia’ to ensure relevant articles were not missed. We also used descriptors of the search terms, such as ‘energy-poor’ and ‘electricity poor’ as proxies for ‘energy poverty’, and descriptors of the country or territory of interest, such as “Chinese” and “Lao”, as well as cities known to be in those locations, such as Seoul, were also included. We only included English-language research and review articles in which the search terms appeared in the title, abstract, or keywords. We excluded articles about national energy security or included a region of interest within a wider scope of countries.

We conducted our database search from 9 to 15 March 2023. We downloaded the metadata of the articles we found into Ris format and imported them into VOSviewer version 1.6.18, where we performed our bibliometric analysis. Bibliometric analysis, also known as bibliometrics, is a method to analyse a large volume of information in an objective and structured manner to identify trends over time, prominent scholars, and present an overview of the existing understanding in the field [35]. This method is particularly advantageous in fields where empirical contributions are voluminous and fragmented [36].

In this article, we used bibliometrics to draw connections between researchers/authors and identify common and emerging themes via keyword searches of article abstracts. We indexed keywords that occurred a significant number of times and excluded irrelevant words (e. g., “attention”, “article”, “percent”). We also included authors who had published at least two articles related to energy poverty to identify emerging scholars. We gained insight into understanding energy-related

deprivations in each country or territory by thoroughly reading the articles. We analysed this corpus for their focuses, definitions of energy poverty, measurements of energy poverty, and culture-specific and general findings.

3. Results

Our corpus comprises 256 relevant articles published between 2004 and 2023 (Fig. 1). The most significant number of articles was on China (89), followed by India (58), Bangladesh (16), Indonesia (15), Pakistan (13), and Japan (12). There were no papers that focused solely on Macau SAR, Singapore, Bhutan, Maldives, and Thailand, although energy poverty in the latter three countries was mentioned in studies on South Asia as a region (see [37–40]). The earliest articles about energy poverty published prior to 2010, were focused on India and Pakistan. Since 2010, research on the topic in Asia has increased, with a surge of interest in 2019. Most of the research on energy poverty in China was published from 2021 to 2023, with the number of published articles each year exceeding the previous year.

We encountered several articles that fell on the periphery of our search scope but still warranted inclusion. For instance, one paper [41] featured ‘energy security’ in its title despite the absence of any search terms specified in Section 2. However, the same article framed energy security as a household-level issue of insecurity rather than the conventional understanding of energy insecurity at the national level. Given this definition’s similarity to interpretations of energy poverty, we opted to include this article in our review. Another paper [42] solely mentioned ‘poverty’ in its title, though the abstract expanded this abbreviation to explicitly refer to ‘energy poverty.’ Consequently, we deemed it appropriate to include this article in our review.

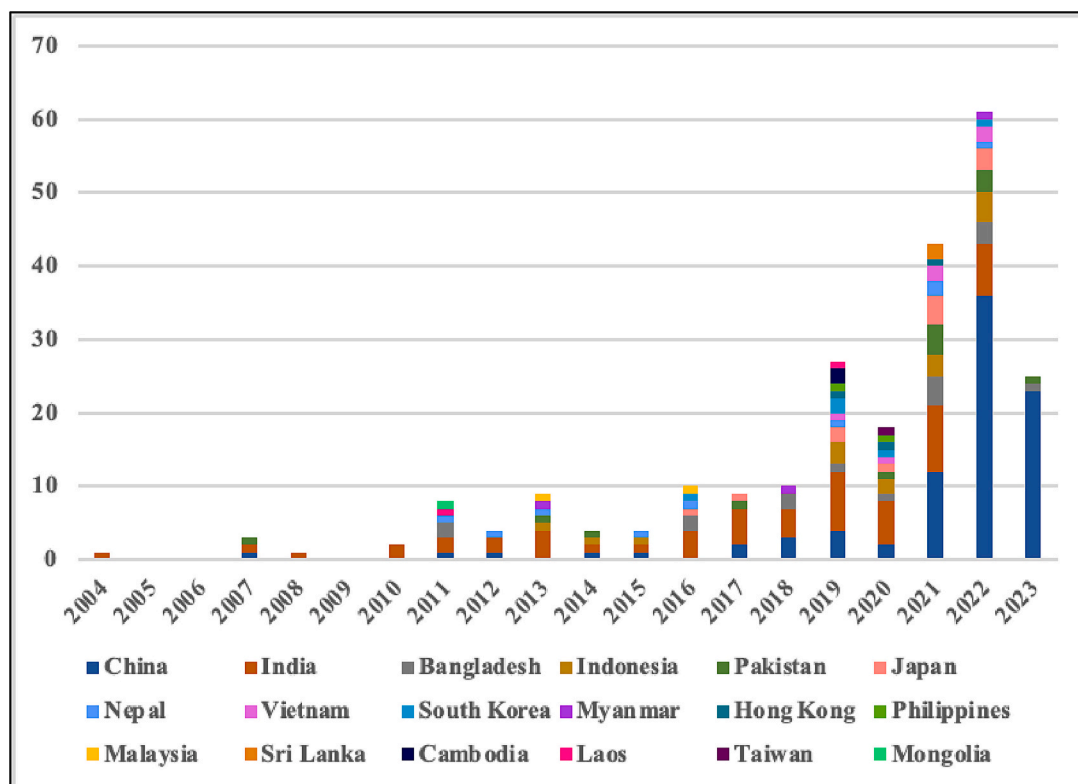


Fig. 1. Geographical distribution of studies on energy poverty in select East, South, and Southeast Asian countries and territories, 2004–2023 (excluding cross-country analyses).

3.1. Bibliometric analysis

Among the 657 authors in our corpus, 96 have published two or more articles on energy poverty in Asia (Fig. 2). We chose the criterion of two or more articles as it reflects an author's genuine interest in the field, rather than being the result of a one-time collaboration. However, our bibliometric analysis reveals that these researchers have few connections or collaborations, indicating a fragmented effort to understand the issue. The network also shows distinct clusters of authors specialise in one particular region. For example, Sadath and Acharya (two green dots in the upper-right corner of Fig. 2; [43,44]) have focused on India; Phoumin and Kimura (two blue dots on the lower-right corner of Fig. 2; [45,46]) collaborated on two studies on Cambodia; Barber and Mah (two blue dots in the lower-left corner of Fig. 2; [47,48]) worked on Hong Kong; while Awan, Sher, and Abbas (three purple dots top-centre of Fig. 2; [49,50]) concentrated on Pakistan, as did Mirza and Qurat-ul-Ann (two green dots on the left of Fig. 2; [51,52]). Phoumin's [53] work outside of Cambodia, where he is based, was about energy poverty in China, but this is excluded from Fig. 2 because none of his co-authors has yet published a second study on the topic. The cluster including Khizar (green, centre of Fig. 2) has focused on South and Southeast Asia [37,54–56].

The bibliometric network not only reveals focused clusters but also identifies established and emerging researchers of energy poverty. In Fig. 2, larger dots indicate more articles published, while darker purple dots indicate earlier authorship, and lighter yellow dots indicate more recent publications.

Three of the earliest authors on energy poverty in Asia – Barnes,

Khandker, and Samad (three purple dots in the center-right of Fig. 2) – conducted research on this topic in Bangladesh in 2011 [57] and India in 2012 [58]. These three World Bank-affiliated authors had their research funded by the Bank's Energy Sector Management Assistance Program (ESMAP). Another early researcher in the field is Sovacool (center upper-left), who published on multiple countries such as Mongolia, Indonesia, Sri Lanka, and India. Sovacool is also part of the largest cluster of authors in the network, which is a linkage of ten authors. However, except for Sovacool, D'Agostino, Bambawale, and Malakar, the other authors in this cluster focus solely on India. Finally, Urpelainen (blue dot near center of Fig. 2) is another expert researcher of energy poverty who has authored 14 articles in the corpus. Unlike the aforementioned researchers who studied multiple countries, Urpelainen has concentrated solely on India.

The bibliometric analysis also revealed clusters of emerging authors in yellow-green clusters. These clusters include Dong, Jiang, Ren, Shahbaz, and Zhao (centre-left), Chang, Gao, Shi, and Xu (upper-right centre), Cheng, Li, Russell, Tani, and Yao (lower-left centre), and Hao, Li, and Zhang (lower right). Notably, these clusters of authors focus exclusively on China, as do Feng (upper-right centre) and Wu (lower-right corner). Munir, on the other hand, stands out as a researcher who has worked on both Bangladesh and China, a rare instance of studying two countries with vastly different geography and culture. Additionally, Das (bottom right corner) has conducted research on both Nepal and India.

We observe that most of the earlier energy poverty researchers are affiliated with universities or organisations in the United States, Australia, and Singapore. Conversely, authors studying energy poverty

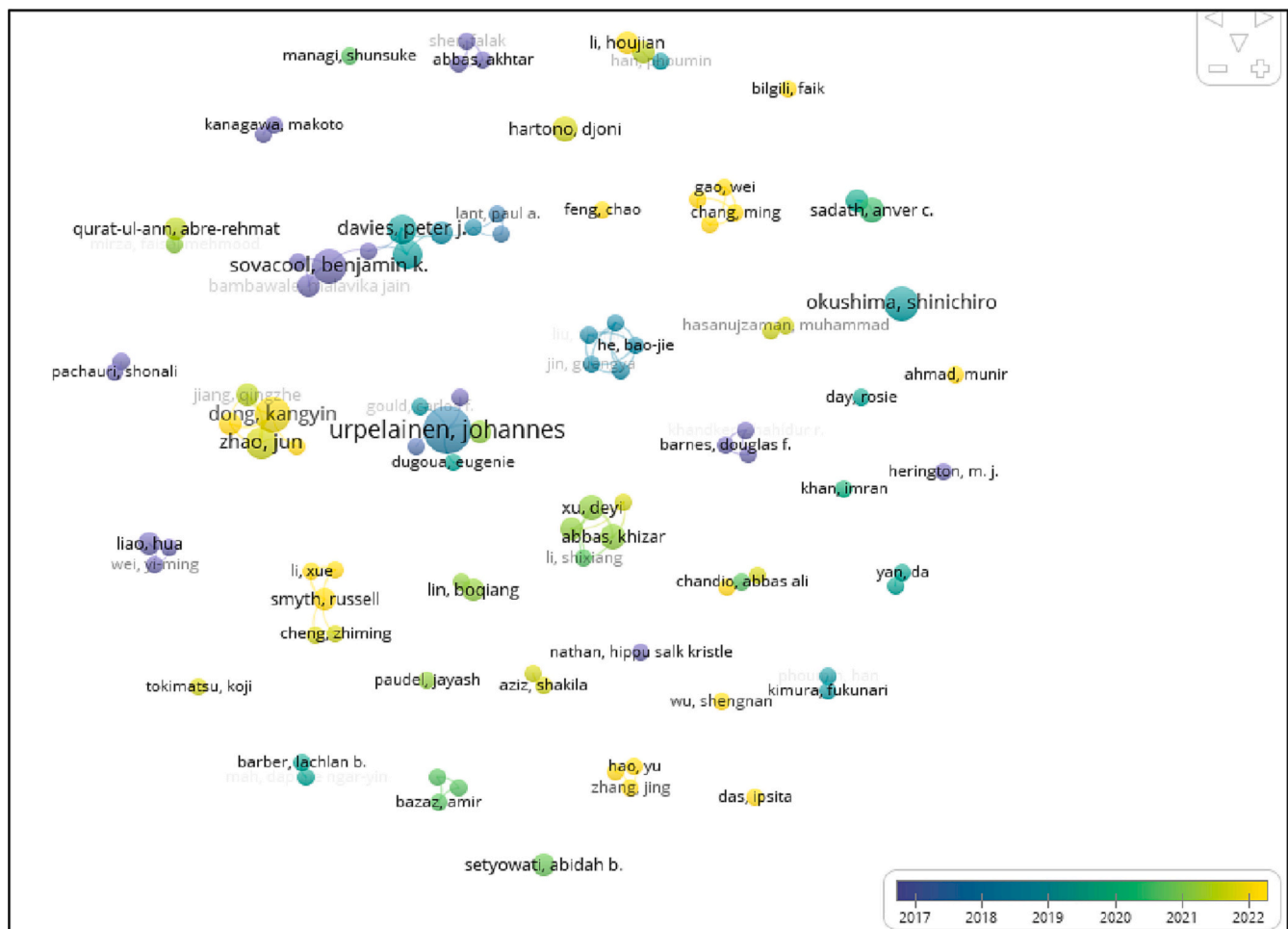


Fig. 2. Network of authors overlaid with year.

in China, Indonesia, and Japan are affiliated with universities within their respective region of study. The former group of researchers mostly conducted studies using national statistics or analysing policies and programs related to energy poverty, except for Urpelainen's extensive work on urban street vendors. While these studies provide valuable insight into energy poverty, there is a noticeable dearth of contributions from researchers within the Global South. Considering that they are situated within the region of interest, their input would be crucial in gaining a deeper understanding of the nuanced aspects of energy poverty that may not be captured by policies and programs alone. In contrast, recent authors of energy poverty studies are predominantly from the Global South (indicated by the yellow to green dots in Fig. 2), and we encourage this trend by recommending further explorations of the lived experiences of energy poverty in this world region. Having a knowledge of the local language would be advantageous in conducting surveys and interviews to uncover local, national, and regional insights that may not be captured through statistical analysis alone. This holistic understanding of energy poverty would facilitate the implementation of more appropriate policies.

The keyword network (Fig. 3) indicates no discernible trends at first glance. However, upon closer examination, there appears to be a possible shift in focus from 'energy access' (blue, center) to 'affordability' (green, upper left corner). The size of the keywords 'China' and 'India' reflects the prevalence of articles in our corpus written about these two countries. Keywords such as 'fuel' and 'electricity' are also

relatively large, indicating a balance between research on electricity (access) and research on (clean) fuels.

The country-related keywords presented in Fig. 4 shed light on the primary research directions pursued in the six countries that have produced the most energy poverty research. These countries are China, India, Japan, Pakistan, Bangladesh, and Indonesia. Notably, for China and India, there is a further distinction made with regard to rural areas. The linkages between pairs of countries, such as Pakistan and Indonesia or Bangladesh and India, suggest that there have been studies comparing or referencing energy poverty in these regions. Interestingly, every region is linked with the keyword 'affordability,' which indicates that financial factors are a common driver of energy deprivation across different energy sources. The keywords related to China, Pakistan, Bangladesh, and Indonesia indicate that achieving SDG 7, which is access to clean and affordable energy, is a key focus for these nations. Specifically, the keywords related to China, both rural and overall, have been linked to the 'China Family Panel Study,' suggesting that most studies rely on quantitative data and use the country's national statistics. The network of keywords related to rural China indicates a current emphasis on 'clean energy' rather than 'cooking' and 'fuel' compared to earlier years. This suggests that there may be limited progress in the access to modern fuels for rural households in China, although research has shifted to a more general focus on fuels rather than specific energy services like cooking.

Pakistan stands out as the only country without a link to

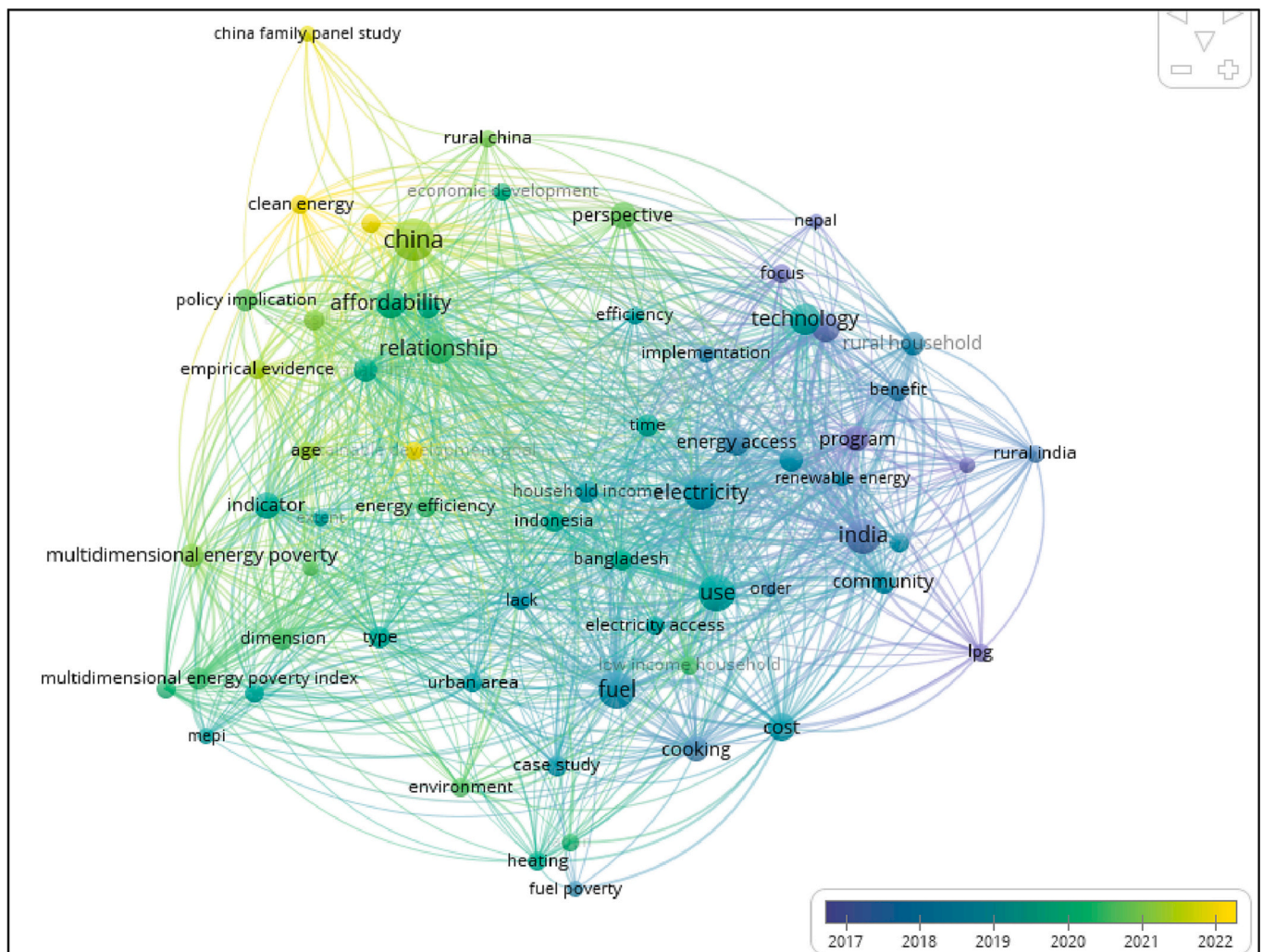


Fig. 3. Network of relevant keywords overlaid with the year 2017–2022.

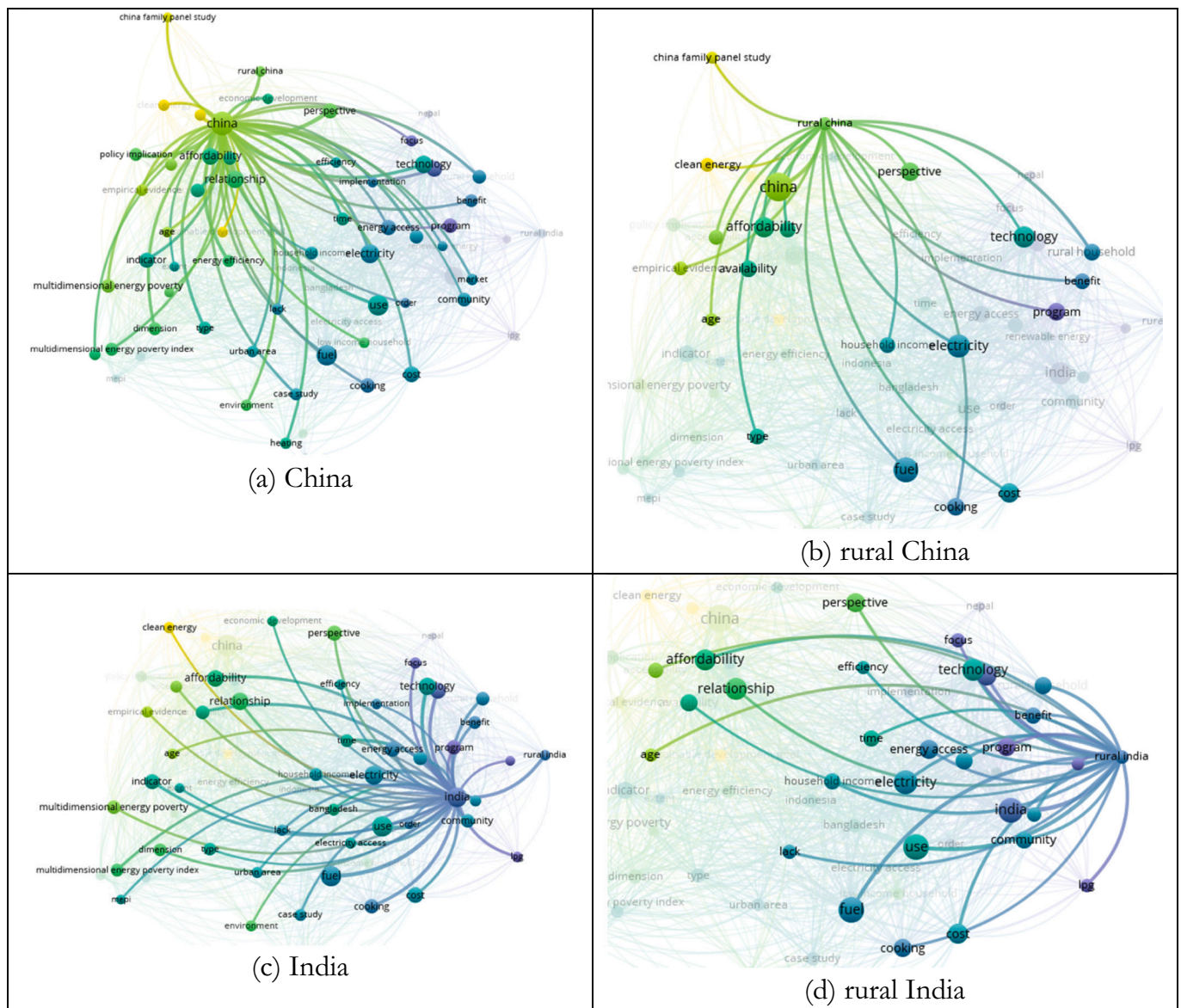


Fig. 4. Keywords associated with China (a), rural China (b), India (c), rural India (d), Japan (e), Pakistan (f), Bangladesh (g), and Indonesia (h). For the legend of colors, refer to Fig. 3.

‘accessibility’, which is unexpected given its status as a developing nation. It is possible that research on Pakistan focuses specifically on the impacts of a particular type of deprivation, such as ‘cooking’ or ‘heating’, rather than investigating factors affecting energy access.

Interestingly, half of the regions examined in this review are associated with ‘heating’, while half are not. Indonesia, India, rural India, and rural China are not linked to ‘heating’. This is surprising, as one might expect ‘heating’ to be associated with rural China due to its cooler climate in the north. Research on energy poverty in Japan focuses on the inability to afford ‘heating’, whereas for countries like Bangladesh and Pakistan, ‘heating’ is an example of an energy service that some households may be unable to provide. This is further supported by their connection to ‘cooking’, which is notably absent in the case of Japan. Keywords related to Japan are linked with ‘energy efficiency’, ‘cost’, ‘accessibility’, and ‘affordability’, suggesting that energy poverty is construed as an inability to afford heating, similar to concepts in developing countries. It is worth noting that Japan is the only country not linked to ‘multidimensional energy poverty’.

3.2. In-depth analysis

Many articles have focused on the factors, impacts, and measurements of energy poverty. In addition, 27 articles investigated the rate of energy poverty within one specific area within a country or territory (Table 1), 21 analysed the use of solid fuels as one dimension of energy poverty, 12 specifically analysed electricity access, and 41 conducted analyses on programs, policies, and technologies aimed at alleviating energy poverty (Table 2).

4. Discussion

4.1. Definitions of energy poverty

There are three approaches to understanding energy poverty: the humanistic approach, the household actions approach, and emerging interpretations.

Initially, some of the earliest studies on energy poverty in Asia

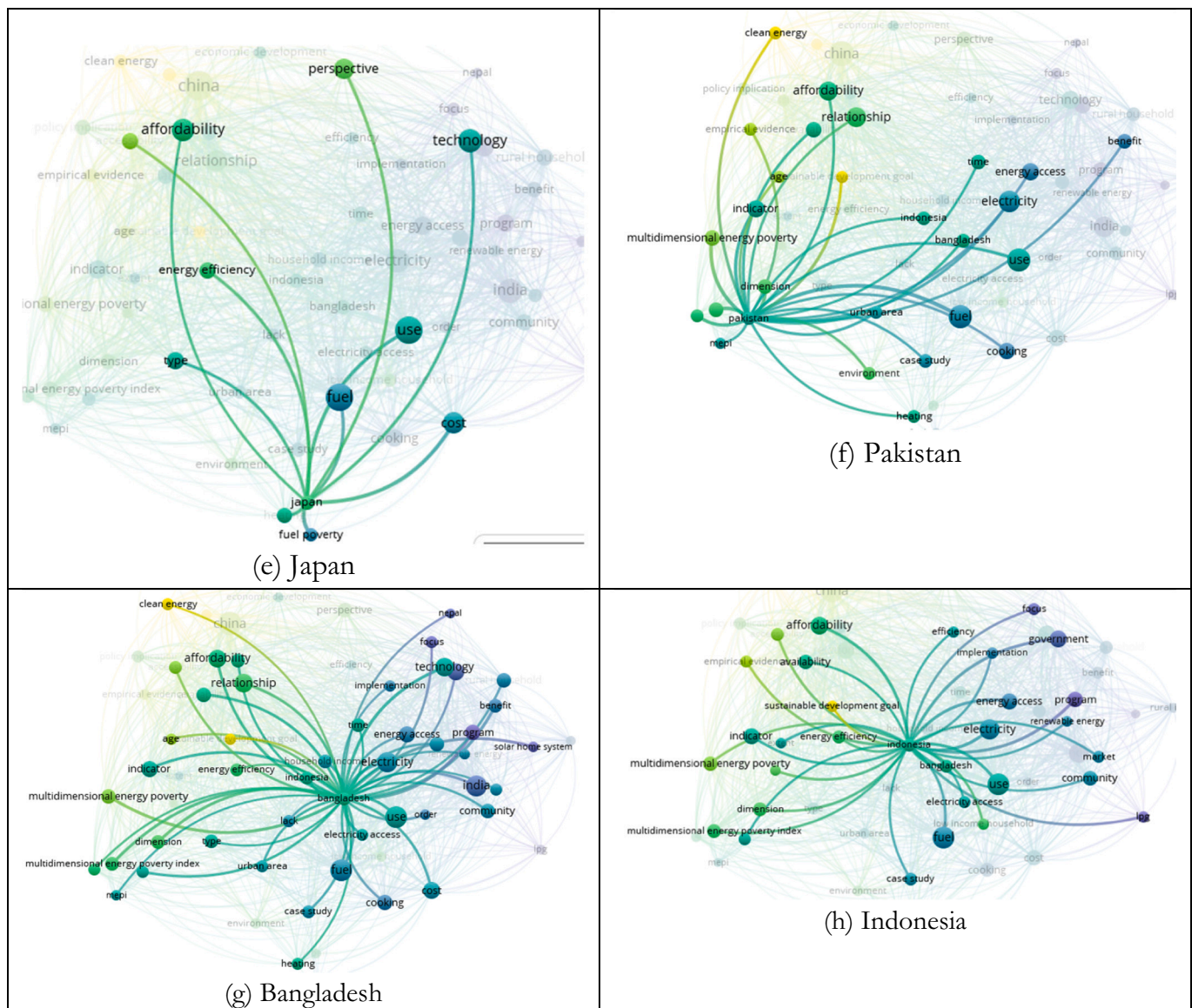


Fig. 4. (continued).

considered energy poverty as a subset of poverty (e.g., [49,50,155–157]). In one of the earliest works on energy poverty in Asia, Barnes et al. defined the issue as ‘the point at which people use the bare minimum energy (derived from all sources) needed to sustain life’, akin to the concept used to define the poverty line [57]. Energy poverty can be understood from the contrapositive perspective, as the inability of households to meet their basic energy needs. This is similar to Bouzarovski et al.’s definition of energy poverty, which emphasises a socially and materially sufficient level of energy services [1]. This definition was commonly cited in our corpus (e.g., [150,152,158–163]). Phoumin and Kimura further developed this concept, coining the term ‘energy insecurity’, which stresses the adequate and sufficient *consumption* level of a household to meet basic energy needs [46]. This highlights the distinction between theory and practice. Regardless of the specific definition, these researchers did not distinguish between ‘fuel poverty’ and ‘energy poverty’, understanding the terms to be interchangeable (e.g., as in [71,150,164]).

However, some researchers continued to make this distinction (e.g., [64,75]). Fuel poverty was defined as the inability to afford clean and modern energy or spending more than 10% of household income on energy, while energy poverty referred to the issue of availability and

accessibility of modern energy services in developing areas. This difference in definition is not surprising, as energy poverty used to be conflated with the concept of ‘energy access’ [165]. In the case of this differentiation, researchers in developing countries, such as Indonesia, understood energy poverty as difficulties in obtaining modern energy services, with some specifying a particular form of modern energy such as electricity (e.g., [124,166]).

Another group of definitions within the same humanistic approach revolves around a household’s right to choose. This was first popularised by Reddy et al. [167], who defined energy poverty as ‘the absence of sufficient choice in accessing adequate, affordable reliable, quality, safe and environmentally benign energy services to support economic and human development’. This concept was predominantly favoured in research on developing countries such as India, Malaysia, Nepal, and Pakistan [49,50,75,106,168].

The second type of understanding was based on the IEA’s definition of energy poverty, which characterised it as the ‘inaccessibility and unavailability of electricity or clean cooking facilities, as well as [a household’s] dependency on traditional biomasses and other solid fuels for cooking’ [60]. Some researchers cited this definition directly (e.g., [60,62,63,169–172]), while others focused exclusively on access to

Table 1
Incidence of energy poverty in a particular country.

Authors	Studied country	Indicator	Year of study	Results
Omar and Hasanuzzaman [59]	Bangladesh	Nussbaumer et al.'s MEPI (modified)	2005, 2010, 2016	Decreased from 53.79% in 2005 and 43.51% in 2010 to 36.33% in 2016
Phoumin and Kimura [46]	Cambodia	10% indicator	2015	27% households energy-poor: specifically, 13% of grid-connected households and 32% of off-grid households
Tang and Liao [60]	China	Use of solid fuels for cooking	2010	75% rural households energy-poor (17.1% and 58.6% households use coal and biomass as their primary cooking fuels, respectively), 8.2% urban households and 36.1% township-area households energy-poor
Lin and Wang [61]	China	10% indicator, LIHC, minimum end-use (MEE)	2014	18.91% households are energy-poor in any one of the three indicators, 13.97%, 10.34%, and 11.41% energy-poor by 10% indicator, LIHC, and MEE, respectively
Zhang et al. [53]	China	10% indicator, if households use solid fuel for cooking	2012, 2014, 2016	2016: 29.44% rural households and 21.4% households exceed energy-poor by 10% indicator; 52.46% rural households energy-poor by using solid fuel for cooking
Huang et al. [62]	China	Nussbaumer et al.'s MEPI (modified)	1989–2015	<i>Discrete values not given, analysed change in severity of energy poverty among households</i>
Lu et al. [63]	China	Self-designed comprehensive measurement	2007–2017	<i>Discrete values not given, analysed change in energy poverty using spatial-temporal analysis</i>
Liang and Asuka [64]	China	Self-designed composite index	2014–2019	Average energy expenditure as a share of disposable income, from 2014 to 2019: 3.18% to 2.47% for urban households, from 4.19% to 3.71% for rural households Northern and western areas suffer from energy poverty more than eastern and southern areas
Pachauri et al. [65]	India	Minimum energy needed	1983–2000	<i>Discrete values not given, analysed by proportion of access to specific sources of energy</i>
Sadath and Acharya [43]	India	Nussbaumer et al.'s MEPI (modified)	2011/12	Least households in Delhi and Chandigarh energy-poor, most households in Uttar Pradesh and Madhya Pradesh
Archaya and Sadath [66]	India	Nussbaumer et al.'s MEPI (modified)	2015–2018	Reduced an average of 30% from 2015 to 2018
Mathen and Sadath [67]	India	Nussbaumer et al.'s MEPI (modified)	2022	In Kasargod, Kerala: 1% households suffer from 90% energy poverty, 48.5% households suffer from 30% or 40% energy poverty
Sambodo and Novandra [68]	Indonesia	32.4 kWh/month consumption, energy spending as 10% of total expenditure	2016	~35% households energy-poor by electricity consumption, ~18% households energy-poor by 10% of total expenditure
Hasibuan and Nasrudin [69]	Indonesia	32.4 kWh/month consumption	2018	~27.7% of poor households (bottom decile) are energy-poor, ~4.6% of richest households (top decile) are energy-poor
Tabata and Tsai [70]	Japan	LIHC and 10% indicator	2004	'Summer fuel poverty': 0.93% of the sampled households fuel-poor, 5.53% of income-poor also fuel-poor
Okushima [71]	Japan	10% indicator (modified)	2004–2013	From 3.2% in 2004 to 5.8% in 2013, with adjustment to winter average is 8.4% in 2004 and 15.2% in 2013
Okushima [72]	Japan	Self-developed MEPI with dimensions of energy costs, income, and energy efficiency	2004–2013	From 3.2% in 2004 to 5.3% in 2013, with adjustment to winter average is 5.5% in 2004 and 8.4% in 2013
Castano-Rosa and Okushima [73]	Japan	10% indicator (modified)	2013–2017	2017 winter: 24.5% in Hokkaido, 46% in Okinawa 2017 summer: 3.9% in Hokkaido, 12.4% in Okinawa
Oum [74]	Laos	10% indicator, if off-grid household	2008/09 and 2012/13	~40% of households energy-poor in 2008, fell to ~24% in 2013
Kitchen and O'Reily [75]	Malaysia	Electricity access rate, if spending on energy sources leaves remaining household income below the poverty line of RM830	2010	99.5% households have access to electricity, though only 96.6% have reliable access, most adversely affected are rural Sabah and Sarawak 40.1% of rural households and 15.8% of urban households experienced difficulties affording energy
Parajuli [76]	Nepal	Minimum energy needed, households having end-use devices with efficiency lower than 25%	2006	57.4% households in total: 52.2% households in urban settlements and 58.6% in rural locations, percentage much higher in mountain regions as compared to hills and terai
Awan et al. [49]	Pakistan	Nussbaumer et al.'s MEPI	2007/08	54.6% overall households energy-poor in 2 dimensions or more, 29% urban households energy-poor, while 71.35 rural households energy-poor
Sher et al. [50]	Pakistan	Nussbaumer et al.'s MEPI	2007/08	47%, 51%, 69% and 66% households in Punjab, Sindh, Khyber Pakhtoon Khaw, and Baluchistan, respectively are energy-poor
Mahmood and Shah [77]	Pakistan	Nussbaumer et al.'s MEPI (modified)	2008/09, 2010/11	56.2% households energy-poor. Specifically: 19% households in urban areas, 76.2% households in rural areas
Awan et al. [78]	Pakistan	10% indicator, double-median approach	1998–2019	10% indicator: from 19.2% in 1999, 26.7% in 2006, and 25.1% in 2008, to 28.6% in 2019 Double-median: 7.6% in 1999, 8.4% in 2002, and 7.3% in 2008, to 11.9% in 2019
Mendoza et al. [79]	The Philippines	Nussbaumer et al.'s MEPI	2016	69.5% households in energy poverty, Autonomous Region of Muslim Mindanao and Zamboanga Peninsula the most severely impacted
Jayasinghe et al. [80]	Sri Lanka	Nussbaumer et al.'s MEPI (modified)	2016	71.6% households in energy poverty, energy-poor households are not always income-poor

Table 2

Summary of papers that have focused exclusively on one dimension of energy poverty or analysed solutions to alleviate energy poverty.

Aspect	Authors	Year	Research summary
Use of solid fuels/traditional biomass	Spagnoletti and Callaghan [81]	2013	Proposed a multilateral alliance to coordinate energy poverty alleviation initiatives in Asia
	Kumar et al. [82]	2016	Studied economic and social determinants influencing uptake of LPG in rural India
	Baquié and Urpelainen [83]	2017	Analysed household satisfaction with cooking arrangements, to determine obstacles in clean fuel adoption in India
	Grové et al. [84]	2017	Evaluated economic viability of cleaner fuels as a solution for energy poverty in India
	Herington et al. [85]	2017	Investigated motivations and social processes behind adopting clean cooking practices
	Hou et al. [86]	2017	Identified situation and determination of cooking fuel choice in rural areas in China
	Gould and Urpelainen [87]	2018	Investigated the views, use patterns, and perceptions of LPG in rural Indian households
	Malakar et al. [88]	2018	Explored role of social structure in continued use of solid cooking fuels in three villages in rural India
	Astuti et al. [89]	2019	Analysed factors in institutional support and household preferences for adoption of LPG in Indonesia
	Renner et al. [90]	2019	Evaluated the economic viability of cleaner fuels for Indonesia
	Gould et al. [91]	2020	Examined role of education and attitudes towards LPG adoption in two Indian states
	Hou and Urpelainen [92]	2020	Examined relationship between lighting choices (using kerosene) and electricity quality in six Indian states
	Neto-Bradley et al. [93]	2020	Identified groups of households that switched to cleaner cooking stoves and their transition pathways
	Neto-Bradley et al. [94]	2021	Identified barriers to LPG uptake in low-income households in Bangalore, India
	Aziz et al. [95]	2022	Explored socio-economic determinants behind clean cooking in Bangladesh
	Das et al. [96]	2022	Studied household preferences and determinants for primary cooking fuels in fuel stacking behaviour in Nepal
	Li and Li [97]	2022	Investigated impact of religion on solid fuel use in ethnic minority households in rural China
	Sharma and Dash [98]	2022	Investigated factors influencing household's cooking energy preference in India
	Talevi et al. [99]	2022	Analysed household preferences for biogas for cooking in rural India
	Zhang et al. [100]	2022	Investigated impact of solid fuel use on life satisfaction of rural residents in China
	Chen et al. [101]	2023	Investigated process and driving factors of household cooking fuel changes in mountainous areas in rural China
Electricity access	Bambawale et al. [102]	2011	Analysed successes and shortcomings of the Rural Electrification Project Phase 1 in Laos
	Borhanazad et al. [103]	2013	Analysed potential of rural electrification policies in Malaysia
	Rao [104]	2013	Investigated the difference in impact of electricity access on income in urban and rural households in India
	Sovacool [105]	2013	Identified four key energy poverty challenges in Myanmar relating to (1) high rates of poverty, (2) basic, conflicting priorities, (3) lack of resources, and (4) uncoordinated policies, proposed solutions such as energy democratisation and grid interconnection in the Mekong area
	Bridge et al. [106]	2016	Explored the impact of electricity access on quality of life in Nepal
	Dugoua et al. [107]	2017	Investigated trends in electricity access in India
	Kim and Jung [108]	2018	Analysed economic competitiveness of Energy Storage Systems in rural Myanmar
	Heynen et al. [109]	2019	Investigated private firms in India's off-grid sector to understand the barriers and hesitations regarding grid expansion
	Son and Yoon [110]	2020	Investigated household electricity use following national grid expansion in Vietnam
	Aziz and Chowdhury [111]	2021	Analysed factors affecting choice of electricity in off-grid areas in Bangladesh
	Ha-Duong and Nguyen [112]	2021	Estimated impact of electricity poverty reduction on self-satisfaction in Vietnam
	Cahyani et al. [113]	2022	Investigated prevalence and household determinants of electricity consumption inequality (insufficiency) in urban and rural Indonesia
Solution to energy poverty — solar home systems (SHS)/solar-related initiatives	D'Agostino et al. [114]	2011	Analysed post-project impacts of Renewable Energy Development Project for SHS deployment in China
	Sovacool et al. [115]	2011	Analysed benefits and remaining challenges of SHS program in Mongolia
	Sovacool and Drupady [116]	2011	Analysis of benefits brought by Grameen Shakti, a company responsible for installing SHS in Bangladesh
	Choragudi [117]	2013	Analysed adoption pattern of solar off-grid lighting systems in India
	Urpelainen and Yoon [118]	2015	Gauged interest and awareness of SHS in rural India
	Szakonyi and Urpelainen [119]	2016	Identified challenges of introducing solar power for lighting among urban poor in India
	Urpelainen [120]	2016	Gauged perception of solar energy in a marginalised community in India, for creating off-grid solar markets
	Kurata et al. [121]	2018	Analysed factors influencing SHS adoption between households and micro-enterprises in southern Bangladesh
	Liu et al. [122]	2018	Case study of the energy consumption and temperature optimisation of a solar house retrofit in Tibet, China
	Sovacool [123]	2018	Case study of one success and one failure of SHS implementation in Sri Lanka and Indonesia, respectively
	Venkateswaran et al. [124]	2018	Analysed factors influencing adoption and use of solar photovoltaic systems in rural areas in India
	Yadav et al. [125]	2018	Assessed reasons behind slow adoption of SHS in rural areas in India
	Yadav et al. [126]	2018	Analysed reasons behind and proposed solutions to counter slow adoption of SHS in rural areas in India

(continued on next page)

Table 2 (continued)

Aspect	Authors	Year	Research summary
Solution to energy poverty — renewable technologies	Khan [127]	2019	Identified the drivers, enablers, and challenges in prosumerism of an SHS in Bangladesh
	Liu et al. [128]	2019	Investigated the potential of a passive solar heating retrofit in Tibet, China
	Yadav et al. [129]	2019	Explored decentralised solar photovoltaic system in India, with focus on access and equity challenges
	Yadav et al. [130]	2019	Identified determinants to increase adoption of SHS in rural households in India
	Zubi et al. [131]	2019	Technical analysis of SHS as energy poverty alleviation method in developing regions
	Abu Saim and Khan [132]	2021	Analysed the household burdens of SHS in Bangladesh
	Akter and Bagchi [133]	2021	Evaluated the success of SHS adoption in alleviating energy poverty in India. In terms of energy justice, people with lower castes, poor households, and day labourers were still left behind
	Zaman et al. [134]	2021	Investigated influence of borrowing constraints on price of SHS loans for rural Bangladeshi households
	Li et al. [135]	2023	Analysed impact of Photovoltaic Poverty Alleviation Project (PPAP) on energy poverty alleviation in China
	Li et al. [136]	2023	Analysis of conformity tendencies and strong-weak ties in behaviour encouraging solar adoption in rural China
	Bhide and Monroy [137]	2011	Identified issues in promoting renewable energy technologies to alleviate energy poverty in rural India
	Gurung et al. [138]	2012	Evaluated Nepal's policies and subsidies to maximise renewable energy resources in isolated and rural communities
	Luo and Zhang [139]	2012	Analysed the status quo and feasibility of using renewable energy in Tibet, China
	Joshi and Yenneti [140]	2020	Presented case studies of grassroot innovations in off-grid energy projects in India
	Aklin and Urpelainen [141]	2021	Explored uptake of distributed renewable energy and its role in alleviating energy poverty in rural India
Solution to energy poverty — energy poverty programs	Wirawan and Gultom [142]	2021	Investigated role of renewable energy-based rural electrification in alleviating poverty in remote villages in Indonesia
	Ahmad and Wu [143]	2022	Examined factors affecting uptake of household-level biogas plants in Bangladesh
	Tanaka et al. [144]	2022	Evaluated the relationship between use of renewable energy and social issues in urban Japan using the SDG as a framework. Found that decentralised RE systems in dense urban areas may exacerbate energy poverty
	Chen et al. [145]	2023	Investigated best renewable energy options for rural China
	Palit et al. [146]	2013	Analysed India's Village Energy Security Programme
	Andadari et al. [147]	2014	Evaluated success of the Indonesian LPG program to help phase out kerosene for cooking
	Jin [148]	2020	Rebound effect of South Korea's fuel poverty program is more pronounced in low-income households
	Lee and Shepley [149]	2020	Assessed effectiveness of South Korean government's energy poverty program of providing solar PV systems
	Setyowati [150]	2020	Difficulties in mobilisation of private climate finance for renewable rural electrification in Indonesia
	Fathoni et al. [151]	2021	Case study of how socio-historical factors shape energy injustices in two Indonesian islands
	Setyowati [152]	2021	Energy justice in Indonesia's efforts to alleviate energy poverty via mobilising private finance for renewable rural electrification
	Shin et al. [153]	2022	Analysis of energy savings after South Korea's energy welfare program of subsidised retrofit of low-income households
	Xie et al. [154]	2022	Identified groups who were negatively affected by China's household clean heating program

electricity (see Table 2), reliance on traditional fuels (e.g., [173,174] and Table 2), or broadened the scope to include other energy services (e.g., [59,127,173,175–178]). This definition was typically used in studies that investigated the accessibility dimension of energy poverty, particularly in rural and developing areas.

Some researchers have proposed their own interpretations of energy poverty. In China, for instance, multiple studies on energy poverty referred to Wang et al. [179], who identified four concurrent definitions of energy poverty: limited access to energy services, failure to meet energy requirements, inability to afford energy services, and difficulties in using clean and advanced energy. Interestingly, these four definitions correspond to the four components frequently identified in energy poverty indicators [179].

Our review identified new “forms” of energy poverty, including ‘hidden energy poverty’ of both clean and non-clean fuels (e.g., [75,180–182]), ‘urban energy poverty’ to describe the inability of urban households to access or afford modern and clean fuels (e.g., [183,184]), and ‘cooling poverty’ for discussing cooling needs within energy poverty (e.g., [185]). Additionally, many studies on India (e.g., [43,44,65,84,186]) called for the inclusion of Amartya Sen’s capability approach. These studies argued that energy poverty was crucial because injustices in energy consumption would have knock-on effects on health and education. By providing access to energy, people would have the opportunity to develop. This idea was also implied in an earlier study

conducted on energy poverty in Indonesia [187]. Furthermore, Yadav et al.’s group (e.g., [125,126,129,130]) proposed the ‘energy poverty cycle’ to emphasise the lock-in effects of the inability to access and afford clean energy.

4.2. Measurements of energy poverty

We found a clear divide in the measurements used between studies that focused on accessibility and those that focused on affordability. However, in both cases, the majority of studies relied on objective, quantitative indicators, while only a few used subjective indicators.

The earliest attempts to measure energy poverty centered on accessibility. For instance, electricity access was used as a proxy to gauge energy poverty (such as [38,39]), while attempts to establish a new energy poverty line borrowed concepts from the poverty line (e.g., [57,58]). However, the use of electrification statistics as an indicator of energy access was challenged by some researchers (as in [75,183,188]), who argued that there was a difference between being connected and receiving an affordable, reliable, sustainable supply. This difference was also highlighted by Aklin et al.’s [189] investigation into the satisfaction of electricity supply over time. As accessibility became secure, people became more concerned about the reliability of the electricity supply. This appears to reflect different “stages” of energy poverty: accessibility, reliability, and potentially efficiency or affordability. To address these

different aspects of energy poverty, separate indicators were proposed, such as Dugoua et al.'s indicator for reliability of electricity access [190] or the threshold of electricity consumption favoured in studies on Indonesia (e.g., [68,69,191]). While focusing on one specific issue or one specific form of energy is not ideal, it may sometimes be necessary due to the unavailability of other data.

As more data become available, a more accurate understanding of the prevalence of energy poverty in a given location may be possible. One tool that has been favoured in exploring access to energy services is Nussbaumer et al.'s MEPI, which offers a comprehensive evaluation of "essential" energy services and can account for various dimensions of energy deprivation. Many studies have either used this indicator directly (see Table 1) or modified the weight ratios and indicators to suit conditions in their country of interest (e.g., [37,79]). However, the weight ratios for each dimension were sometimes arbitrarily assigned, reducing the significance of the final numerical output of the MEPI.

The concept of multidimensionality was a central consideration for many new composite indicators that measure energy accessibility. The addition of new and unique components also reflected the regional context for which the indicators were proposed. For example, Lozano and Taboada's indicator included a component to assess the legality of electricity contracts, suggesting that legal structure is important in discussing the state of Philippine energy poverty [192]. The inclusion of geographical characteristics was also prevalent in indices suggested by researchers studying India. Yadava and Sinha attempted to design an indicator specific to the Indian forest fringe villages [193], while Gupta et al. included forest area, which was hypothesised to indicate easier access to traditional fuels [194]. Researchers also moved beyond strictly assessing the energy system in terms of accessibility, affordability, efficiency, or even the renewable energy component of the energy mix, to include ideas such as inequality via the inclusion of the Gini coefficient (e.g., [195]), or the role of investments and technologies (e.g., [63]). Some also attempted to establish a continuum of energy poverty severity instead of taking a binary approach (e.g., [65,67,194,196]).

It appears that the concept of affordability in energy poverty measurement has been mainly focused on the 10% indicator proposed by Boardman, which is commonly used in more developed areas such as Japan, Korea, and Hong Kong (e.g., [47,71,181,184]). However, in developing areas like Indonesia, LPG has been the focus of affordability measurement [197]. Several studies have made reference to the Low Income High Cost (LIHC) indicator benchmarked against the median expenditure (e.g., [198]), while others used the actual energy consumption (e.g., [199]). In terms of multidimensionality, there have been attempts to combine statistics of energy consumption and spending, utilising new methods. For example, Okushima proposed an MEP for developed countries with regional variations in climate and considered energy costs, income, and energy efficiency of housing [72]. Similarly, Jiang et al. proposed a new method for determining the energy poverty line based on the point at which an increase in income no longer has a significant impact on energy consumption [169]. The energy poverty comprehensive index proposed by Wang et al. [179], commonly cited in studies investigating energy poverty in China (e.g., [200–203]), is also noteworthy. It assesses both accessibility and affordability of energy, encompassing 23 measurements across four aspects: energy service availability, energy consumption cleanliness, energy management completeness, and household energy affordability and efficiency.

In the traditional dichotomy between accessibility and affordability, we found that developed countries tended to focus on affordability, particularly in relation to electricity. Developing countries, such as Bangladesh, Nepal, Pakistan, and Sri Lanka, on the other hand, tended to measure energy poverty in terms of accessing clean, modern fuels or energy services. However, studies in some developing regions, such as China, India, and Cambodia, acknowledged that the issue of energy poverty in their respective countries exhibited features of both accessibility and affordability (e.g., [53,61,64,179]). This is an important development as it suggests that future research may be able to bridge the

conceptualisations of energy poverty between the Global North and the Global South.

Moving away from objective, quantitative measurements, we note that only a handful of studies have used subjective indicators of energy poverty via questionnaires or interviews (e.g., [48,175,180,182,183,204,205]). Instead, many studies have used national-level statistics to investigate the incidence and intensity of energy poverty. While this is useful to approximate the number of households in energy poverty, the determinants of energy poverty through these studies only describe the probability of certain households being energy poor. As such, policy recommendations based on these findings may only be able to alleviate energy poverty for a fraction of households. Subjective studies of energy poverty experiences are therefore needed to fully understand the issue, and formulate targeted policies based on lived experiences.

One recurring critique in the literature (e.g., in [57,164]) is the lack of a precise and consistent method for measuring energy poverty. Commonly used indicators, such as the 10% indicator and the MEPI, have been deemed insufficient in accurately measuring energy poverty due to their inability to account for local conditions. Consequently, some researchers have proposed composite and contextualised measures of energy poverty, resulting in a proliferation of energy poverty indicators. This multiplicity of indicators makes comparisons between countries challenging and leads to fragmented research on energy poverty.

To address this issue, we propose a more general definition of energy poverty and a measurement framework that can be customised to specific study areas. This approach is akin to the use of Nussbaumer et al.'s MEPI, which features several categories of energy deprivations with different indicators and corresponding weight ratios that researchers can modify. This method can also be complemented by the "continuum approach", which involves grouping households that fall into similar percentages of deprivation. Evaluating the incidence of households in these categories may be more meaningful than examining the numerical index directly.

4.3. Variables affecting energy poverty

This section discusses three sets of variables affecting energy poverty. The first set is composed of micro-level socio-economic variables that are considered factors or determinants that increase the likelihood of a household experiencing energy poverty. The second set consists of macro-level variables, such as urbanisation and warmer temperatures, which have an impact on the overall energy poverty rate. The third set extends the previous two by incorporating an idea similar to Thomson et al.'s energy vulnerability framework [9]. The framework describes six explicit factors (accessibility, affordability, flexibility, energy efficiency, needs, and practices) that influence a household's vulnerability to energy poverty.

Common socio-economic variables associated with energy poverty in households seeking clean fuels include a male-headed household [51,176], a less educated household head [51,176], and lower-income [154,176,181,206]. Working in the agricultural sector is also a factor in some cases [207,208], as is proximity to a district border in Bangladesh [176] and higher latitude in Pakistan [52]. On the other hand, higher levels of development, urbanisation, and education have been found to be associated with households that are not energy-poor [44,172,176,209]. However, in Tibet, energy poverty was not found to be related to age, gender, or education status but instead correlated with household income, employment, and altitude [170]. Furthermore, urbanisation does not guarantee that energy needs are met, as low-income urban households were found to be energy-poor in Laos and South Korea [74,181], with increased vulnerability if the household was female-headed [113]. The variable of age had different impacts depending on the dimension of energy poverty. In terms of accessing clean fuels, households with older heads are less likely to be energy and income poor, likely due to wealth accumulation over time [37,42].

However, in the construct of affording energy, the elderly population may be at higher risk of energy poverty because they are more likely to use energy-inefficient appliances, which increases their energy consumption [73]. External factors can also have an impact on energy poverty, such as remittances [177], population migration [210], degree of ethnic fractionalisation [211], presence of forests [212,213], and, of course, warmer temperatures [198,214–216]. Increased temperatures lead to increased needs for cooling, which more people may be unable to afford.

Studies in China have examined various factors and their effects on energy poverty. Some have looked at the impact of the energy transition on energy poverty (e.g., [217,218]), while others have attempted to identify factors that could alleviate it (e.g., [219]). Many of these studies have focused on digital aspects. It has been found that increasing fiscal decentralisation [200,220], promoting renewable energy technology innovation [221,222], improving the digital economy [201], increasing mobile phone use [223], promoting financial inclusion [224], reducing the digital divide [225], increasing economic growth [226], and increasing social capital [227] could all help alleviate energy poverty. It should be noted that increased use of mobile phones does not directly alleviate energy poverty, but it can lead to intermediate pathways, such as increased awareness, better infrastructure, and more innovation, which can help people improve their situation. Other studies have looked at inter-provincial energy exportation, digital finance, and green finance potential solutions to energy poverty [171,228–230]. In South Asia, the impact of financial inclusion on energy poverty has also been explored [231].

One discussion that deserves attention is the promotion of clean energy and its potential impact on energy poverty. At first glance, promoting clean energy seems like a logical step to alleviate energy poverty since the core problem is the inaccessibility or unaffordability of clean fuels. However, there are contrasting views in this regard. Hong et al. [232] found that promoting clean energy could be a significant help, particularly for women and people with lower levels of education. Conversely, Xie et al. [154] argued that promoting clean energy could inadvertently worsen energy poverty. This is because the cost of cleaner fuels is generally more expensive, which could impose an additional burden on low-income households and households with elderly members. Although these two studies focus on China, this discussion could be relevant to other regions as well. Policymakers must be cautious and carefully evaluate the implications of promoting clean energy. Similar to Thomson et al.'s [9] energy vulnerability framework, some studies have focused on exploring the determinants that put a household or person at risk of experiencing energy poverty. These determinants include physical factors such as building structure and energy efficiency (e.g., [47,163,182,233,234]), climatic factors such as the region's climate and increased occurrences of extreme weather (e.g., [182,235]), socio-economic factors like landlord-tenant relations (e.g., [47]), and cultural factors like bathing practices, gift-giving, and religion (e.g., [73,236,237]). For instance, increased hot weather would impact a household's energy needs, especially for cooling [162]. Taking this a step further, studies within a particular region also account for local events. For example, Herington and Malakar [238] situated their research in the wake of the 2015 Nepal humanitarian crises, Okushima [71,72] noted the impact of the 2011 Great Eastern Japanese Earthquake on energy poverty in Japan, and Cheng et al. [239] contextualised their findings on childhood adversity and the probability of experiencing energy poverty in later life using the Great Chinese Famine.

4.4. Impacts of energy poverty

The discussion in this section centres around three main focuses. The first pertains to micro-level impacts that directly affect households, such as education, health, or depression. The second focus is on macro-level impacts, including electricity demand and carbon emissions. The third builds upon the first by examining the household's response or coping

behaviour towards energy poverty. This is a critical consideration because it views those experiencing energy poverty as active agents capable of overcoming this deprivation. It is worth noting that although insights into coping behaviour are valuable, obtaining data on this topic is challenging, as it requires qualitative analysis and subjective indicators of energy poverty.

Firstly, studies have emphasised the negative impact of energy poverty on education, health and wellbeing, and quality of life (e.g., [45,53,56,59,235,240–248]). Energy poverty, especially in developing regions, impedes children's education as they spend time collecting fuel instead of studying [45,59]. During the COVID-19 pandemic, the impact was compounded as some children lacked access to computers and phones required for online education [80]. Conversely, improving access to electricity was modelled to improve literacy rates [249]. Moreover, energy poverty also affects the proportion of household expenditure. For instance, Sambodo and Novandra found that non-energy-poor families in Indonesia spent approximately 16.2% more on food and 24.3% more on non-food purchases than energy-poor families [68]. Similarly, Phoumin and Kimura found that in Cambodia, for every 1% increase in the share of energy expenditure to total expenditure, the household's food consumption would decrease by 4.1% [46]. Thakur found that energy-poor Indian households spent more on health and fuel [250]. Besides, established vulnerable groups, such as women [55,251] and the elderly [244,252], were particularly focused on when examining individual-level impacts.

The studies conducted in China uncovered some uncommon outcomes of energy poverty. One study suggested that experiencing hardship due to energy poverty was associated with a higher probability of becoming an entrepreneur [253]. Another study found that energy poverty was linked to depressive feelings, which may discourage individuals from using the Internet [254]. The relationship between energy poverty and the Internet in China is particularly noteworthy as Internet development was previously thought to alleviate energy poverty in a separate study [255].

Secondly, researchers have also been interested in studying the broader impacts of energy poverty. From an environmental perspective, studies have explored the effects of energy poverty on the environment [256–258], such as its impact on CO₂ emissions [202,259,260], agricultural productivity [261], and irrigation efficiency [262]. Conversely, one study investigated the impact of tightened environmental regulations on energy poverty and found that households relying on dirty fuels were adversely affected, with the severity of their energy poverty increasing [263]. Under the theme of sustainability, researchers have investigated the impact of energy poverty on the rebound effect [264], electricity demand [265], emission reduction targets [266], green growth [267], the low-carbon transition [268], and sustainable development [263,267,269]. These studies were mainly conducted in East Asia, particularly in China, Taiwan, and South Korea.

Lastly, this discussion also includes responses to energy poverty. These responses were within studies that focused on vulnerable groups who were particularly affected by energy poverty. Studies in Hong Kong examined households living in sub-divided units and found that they had severely reduced their energy consumption by limiting the use of air conditioning and heated water, or by adopting behaviours to avoid energy consumption, such as leaving home for a cooler place [48,204]. This coping method was also observed among internally displaced people in Nepal [182]. Phoumin and Kimura also suggested that some Cambodian households may be under-consuming electricity to qualify for the government's initiative of free electricity for households consuming below a certain level [45]. Coping methods for energy poverty fall outside of the measurements covered in Section 4.2, emphasising the importance of subjective indicators and the need to understand the experiences of energy poverty through qualitative research approaches.

4.5. Solutions to address energy poverty

Studies that focus exclusively on analysing policies and technologies to alleviate energy poverty are summarised in Table 2. The analysis of these studies indicates that there is no clear one-size-fits-all solution. This fact is also highlighted in many studies that provide recommendations for governments and policymakers based on their research on energy poverty in a given area. They urge policymakers to consider local conditions when proposing programs to alleviate energy poverty. For example, some studies recommend maximising local resources for renewable energy technologies [259] and targeting social, cultural, and economic variables specific to the locality [270]. Other studies noted the difference between households that are energy-poor and those that are income-poor, cautioning that policies must target the correct group [200,271,272].

The studies in Table 2 demonstrate that renewable energy can be an effective solution to alleviate energy poverty in both developing and developed countries (e.g., [38,40,51,73,149,152,199,205,273,274]). Renewable energy technologies have the potential to address issues of the availability of modern energy services, ensure a reliable supply of clean energy, and reduce the cost of electricity, especially for low-income households.

Many studies have analysed the variables and impacts of energy poverty and have suggested actions for local policymakers. For example, given that energy-poor households are more likely to be in rural areas, have low income, or possess low levels of education, some studies proposed governments should target low-income households in rural areas and increase awareness and education. Additionally, many studies have called for government action, such as strengthening rural energy infrastructure [55] and collaborating with international development agencies [258].

One possible avenue for alleviating energy poverty, not mentioned in Table 2, is via remittances. Remittances do not directly address energy poverty in the same way as solar home systems and energy poverty programs, rather, they can increase a household's income and enable them to access energy services and use cleaner sources of energy. Studies conducted in Bangladesh and Pakistan [51,177] have shown that households receiving foreign remittances are less likely to experience energy poverty and are comparatively less deprived than those not receiving remittances. Based on these findings, one recommendation is for policymakers to implement policies that facilitate remittances and direct them towards community-based renewable energy initiatives.

4.6. Changing times and constructs

The distinction between different dimensions of energy poverty, namely that rural and developing regions experience difficulties accessing clean energy while urban and developed regions face challenges in affording electricity or gas, is not as clear-cut as previously imagined. Aung et al.'s study on Myanmar's rapidly urbanising areas found that rural households, on the contrary, enjoyed more hours of electricity availability and more reliable electricity compared to urban households [183]. Similarly, Herington and Makalar [238] found that in Nepal, after the 2015 humanitarian crisis, households with modern cooking stoves reverted to using traditional cooking stoves and biomass due to the rising fuel costs. They also proposed that perhaps those who were traditionally considered energy impoverished, i.e., those using biomass and non-modern fuels, were actually secure in their access to energy. In contrast, the energy non-poor were energy insecure as they could not maintain a stable supply of energy.

Extending this discussion, it is noteworthy that only a small fraction of studies have focused on the relationship between energy poverty and energy injustice or inequality (e.g., [110,112,151,152,175,205,275–279]). Interestingly, some studies in Japan and Vietnam situated energy poverty within the context of the energy transition (e.g., [42,158,205]). Research on energy poverty in Vietnam, after analysing

electricity consumption and income, suggested that reducing energy poverty may cause an increase in energy injustice, as many low-income households may be unable to enjoy the benefits of increased electricity connection [42,110,112].

Despite significant efforts to contextualise energy poverty, and given the rapid growth and urbanisation in Asia, few articles explore energy poverty in densely populated urban areas (e.g., [163,183,265,280]). Studies conducted in Hong Kong, one of the world's densely populated metropolises, focused only on people living in sub-divided units ([48,204]), without considering other vulnerable groups. Studies on energy poverty in Seoul emphasised the city's dense population [149], but only one study explored who could not afford fuel in this metropolitan region [181]. Research on the energy-poor populations in Asian cities is insufficient. Our review revealed that energy poverty is not only experienced by those in rural areas but is increasingly shared by urban households. Thus, more qualitative research should be conducted on urban households, who may not have the option of relying on traditional fuels, unlike rural households.

Based on our observations, we suggest three recommendations. Firstly, we propose the concept of 'energy insecurity' as a multidimensional construct that considers the physical, socio-economic, and climatic vulnerabilities that households face in relation to experiencing energy-related deprivations. We define energy insecurity as 'an inability to adequately meet basic household energy needs' [281]. This concept is similar to the constructs of secure access to energy that were found in some of the articles we reviewed (e.g., [41,238]). A framework for measuring energy insecurity could include indicators of affordability, accessibility, and reliability. By using this framework, one can reexamine the issue of energy transition and its potential impact on increasing energy poverty. For example, if households shift from clean fuels to traditional biomass, they would still be considered energy insecure under our proposed definition.

While our paper has uncovered significant gaps in energy poverty research, we also acknowledge its limitations. Firstly, we limited our search terms to the primary and conventional terms used to describe energy-related challenges when constructing the dataset for our bibliometric analysis, which may have limited the number of papers reviewed. We excluded terms such as 'energy vulnerability', 'energy access', 'energy reliability', or more specific energy difficulties like 'electricity poverty' and 'fuel stacking'. Although these terms were prominent in our review, we contend that our search strings were sufficient to warrant a structured and systematic analysis. Secondly, we excluded conference proceedings and working papers, which may have resulted in missing essential information about a specific country, territory, or region. We chose to narrow our review rather than aiming for comprehensiveness. Thirdly, we only selected articles that explicitly mentioned the country, territory, or region of interest in the title, keywords, or abstract. Therefore, we may have overlooked comparative studies that focused on a particular country in Asia but were not explicitly mentioned in the title. Almost half of our final corpus focused on China and India, heavily skewing our bibliometric maps towards these two countries.

5. Conclusion

In this paper, we conducted a review of the extant literature on energy poverty studies in 23 countries and territories in East, South, and Southeast Asia, using 256 review and research articles published between 2004 and 2023. The majority of these articles were published post-2018.

Our analysis revealed that many developing Asian countries still view energy poverty as a matter of access to electricity or clean fuels, while developed countries in the region concentrate on electricity affordability. Nevertheless, there has been a growing body of research that comprehensively understands energy poverty. Notably, research on energy poverty in Cambodia, China, and Nepal has challenged current definition of energy poverty, proposing the adoption of the energy

insecurity framework that places emphasis on non-socioeconomic factors contributing to energy insecurity.

We found that research in Asia, compared to other world regions, particularly in the Global North, was increasingly tailored to its study area. Because Asia has a vast number of countries and territories, developments were not as consistent as in the Global North. Nevertheless, new ideas and indicators were proposed that accounted for the many facets of energy poverty within the region. This was best exemplified by the abundance of studies conducted in China, which explored the impacts of digital and financial developments on energy poverty, and in India, which delved into the determinants of energy poverty in greater depth. Despite this progress, we believe there is still insufficient consideration of energy poverty in the context of the increased extreme weather resulting from climate change, as well as the rapid urbanisation and economic growth in the region. There is a scarcity of research assessing energy poverty in urban areas, particularly in relation to cooling needs. By adopting a more comprehensive approach that utilises the concept of energy insecurity, several possible future research directions open up, including (1) identifying who in urban areas experiences energy insecurity, (2) investigating the impacts of energy insecurity on social life, health, and well-being, (3) examining the multiple and possibly intersectional vulnerabilities that cause households to increase their risk of energy insecurity, and (4) exploring how households cope. Moreover, moving away from the rural-urban binary, these same questions could also be applied to households living in rapidly urbanising areas.

Another future research focus should be on situating energy poverty within the context of a just energy transition. Although some research has been conducted on energy-poor households' attitudes toward low-carbon transitions and alleviating energy poverty as a means of achieving the transition, there is a need for a deeper understanding of how to achieve it in a just, equitable, and fair manner while also addressing past and present injustices in energy access [282]. One of the essential questions that future research should address is whether it is possible to identify energy-insecure households and ensure their participation in the energy transition.

Declaration of competing interest

We declare no conflict of interest.

Data availability

No data was used for the research described in the article.

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